

Your first fitness number

Six minutes of walking, or twelve of running, on flat ground. Then one line of arithmetic and a table.



1 · PICK YOUR TEST

Walk six minutes as far as you can.
If you already run, run twelve minutes. Flat ground: a track, a park loop, a quiet road.

2 · MEASURE THE DISTANCE

In metres. A 400 m track makes it easy; otherwise a phone map or a watch. Write it down the moment you stop.

3 · FIND YOUR TIER

Turn the distance into METs with the line below, then read your age row. One MET is you at rest; ten is ten times that.

A heart condition, chest pain on exertion, or told not to exert yourself? Walk, do not run, and ask your doctor first.

If you ran twelve minutes

$VO2max = (\text{metres} - 504.9) \div 44.73$
 $METs = VO2max \div 3.5$
2,400 m: $(2400 - 504.9) \div 44.73 = 42.4$; $42.4 \div 3.5 = \mathbf{12.1 METs}$.
Cooper KH, JAMA, 1968.

If you walked six minutes

$VO2max = 61.1 - 11.1 \times \text{sex} - 0.4 \times \text{age} - 0.2 \times \text{weight kg} + 0.02 \times \text{metres}$
sex: 1 for a man, 2 for a woman · $METs = VO2max \div 3.5$
A man of 45, 80 kg, 600 m: $61.1 - 11.1 - 18 - 16 + 12 = 28.1$; $28.1 \div 3.5 = \mathbf{8.0 METs}$.
Hong et al, 2019, young adults. Over 60 the app uses a line that also takes height (Šagát et al, 2023).

Men, METs by age

Age	Low	Below avg	Above avg	High	Elite
20-29	to 9.3	9.4-10.3	10.4-12.0	12.1-13.2	13.3+
30-39	to 8.9	9.0-10.0	10.1-11.6	11.7-12.8	12.9+
40-49	to 7.9	8.0-9.9	10.0-11.9	12.0-13.9	14.0+
50-59	to 6.9	7.0-8.9	9.0-10.9	11.0-12.9	13.0+
60-69	to 5.9	6.0-7.9	8.0-9.9	10.0-11.9	12.0+
70+	to 4.9	5.0-6.9	7.0-8.9	9.0-10.9	11.0+

Women, METs by age

Age	Low	Below avg	Above avg	High	Elite
20-29	to 6.6	6.7-8.2	8.3-9.3	9.4-10.5	10.6+
30-39	to 6.4	6.5-7.6	7.7-8.9	9.0-10.1	10.2+
40-49	to 5.9	6.0-7.9	8.0-9.9	10.0-11.9	12.0+
50-59	to 4.9	5.0-6.9	7.0-8.9	9.0-10.9	11.0+
60-69	to 3.9	4.0-5.9	6.0-7.9	8.0-9.9	10.0+
70+	to 3.4	3.5-4.9	5.0-6.9	7.0-8.9	9.0+

Ages 40 and over: Mandsager K et al., JAMA Network Open, 2018, 122,007 patients at the Cleveland Clinic, treadmill tested. Under 40: Cooper Institute and ACSM norms, 1997, $VO2max \div 3.5$. In the Cleveland Clinic study, below average rather than above carried about the same risk of dying as smoking or diabetes. A tier is a place in a table, and it moves.

Your result. Keep this sheet, and test again.

Date	Walk or run	Metres	VO2max	METs	Tier
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

The number only means something next to the one before it. Test again after about eight weeks of training. Sooner is hard to tell from a good or bad day (Hickson et al., 1981). That is the window the app uses.